

PSET 3: Introduction to Matrices and Vectors

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Sketch a function

Sketch the graph of a function (any function you like, no need to specify a functional form) that is:¹

- Continuous on $[0, 3]$ and has the following properties: an absolute maximum at 0, an absolute minimum at 3, a local maximum at 1 and a local minimum at 2.
- Do the same for another function with the following properties: 4 is a **critical number** (i.e. $f'(x) = 0$ or $f'(x)$ is undefined), but there is no local minimum and no local maximum.

Problem 2: Find critical values

Find the critical values of these functions:²

- $f(x) = 5x^{2/3} - 4x$
- $s(t) = 3t^4 - 4t^3 + 6t^2$
- $f(r) = \frac{r}{r^2 + r + 1}$
- $h(x) = x \ln(x)$

Problem 3: Find absolute minimum/maximum values

Find the absolute minimum and absolute maximum values of the functions on the given interval:³

- $f(x) = 3x^2 - 12x + 5$, $[0, 1]$
- $f(t) = t^2\sqrt{9 - t^2}$, $[-1, 4]$
- $s(x) = x - \ln(x)$, $[1/2, 2]$

¹inspired by Grimmer HW3.1

²inspired by Grimmer HW3.2

³inspired by Grimmer HW3.3

Problem 4: Basic matrix arithmetic

If

$$\mathbf{a} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

find:⁴

- a. $\mathbf{a} - \mathbf{b}$
- b. $-4\mathbf{b}$
- c. $3\mathbf{a} + 3\mathbf{b}$

Problem 5: More complex matrix arithmetic

Suppose

$$\mathbf{x} = \begin{bmatrix} 3 \\ 2q \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} p+2 \\ -5 \\ 3r \end{bmatrix}$$

If $\mathbf{x} = 2\mathbf{y}$, find p, q, r .⁵

Problem 6: Check for linear dependence

Which of the following sets of vectors are linearly dependent?⁶ In each part, you can denote each vector as $\mathbf{a}, \mathbf{b}, \mathbf{c}$ respectively.

a. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$

b. $\begin{bmatrix} 13 \\ 7 \\ 9 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 5 \\ 8 \end{bmatrix}$

c. $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$

Problem 7: Vector length

Find the length of the following vectors:⁷

- a. $(4, 2)$

⁴inspired by Pemberton and Rau 11.1.2

⁵inspired by Pemberton and Rau 11.1.3

⁶inspired by Pemberton and Rau 11.1.4

⁷inspired by Simon and Blume 10.10

- b. $(2, 2, 2)$
- c. $(4, 2, 3, 1)$
- d. $(0, 0, 0, 0, 3)$

Problem 8: Law of cosines

The **law of cosines** states:

$$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}$$

where θ is the angle from $\mathbf{w}$ to $\mathbf{v}$ measured in radians. Of importance, $\arccos()$ is the inverse of $\cos()$:

$$\theta = \arccos\left(\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}\right)$$

For each of the following pairs of vectors, calculate the angle between them. Report your answers in both radians and degrees. To convert between radians and degrees:⁸

$$\text{Degrees} = \text{Radians} \times \frac{180^\circ}{\pi}$$

- a. $\mathbf{v} = (1, 0), \quad \mathbf{w} = (2, 2)$
- b. $\mathbf{v} = (4, 1), \quad \mathbf{w} = (2, -8)$
- c. $\mathbf{v} = (1, 1, 0), \quad \mathbf{w} = (1, 2, 2)$

Problem 9: Matrix algebra

Using the matrices below, calculate the following. Some may not be defined; if that is the case, say so.⁹

$$\mathbf{A} = \begin{bmatrix} 3 \\ -2 \\ 9 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 8 \\ 0 \\ -1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 7 & -1 & 5 \\ 0 & 2 & -4 \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 3 & 1 \\ 3 & 4 \\ 3 & -7 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 0 & -4 \\ -2 & 1 & -6 \end{bmatrix}$$

$$\mathbf{F} = \begin{bmatrix} 4 & 1 & -5 \\ 0 & 7 & 7 \\ 2 & -3 & 0 \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} 2 & -8 & -5 \\ -3 & 7 & -4 \\ 1 & 0 & 3 \\ 1 & 2 & 6 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} 9 \\ -2 \\ -1 \\ 0 \end{bmatrix} \quad \mathbf{L} = [5 \quad 0 \quad 3 \quad 1] \quad \mathbf{M} = \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}$$

- a. $\mathbf{A} + \mathbf{B}$
- b. $-\mathbf{G}$
- c. $\mathbf{D}'$
- d. $\mathbf{C} + \mathbf{D}$

⁸inspired by Simon and Blume 10.12

⁹inspired by Grimmer HW5.3

- e. $\mathbf{A}'\mathbf{B}$
- f. $\mathbf{BC}$
- g. $\mathbf{FB}$
- h. $\mathbf{E} - 5\mathbf{I}_3$
- i. $\mathbf{M}^2$

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.